N. Akshaya
192425129

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **CSA5101**

Course Name: **Cryptography and Network Security**

Duration: **1: 30 Hr**

Date :

50 Marks

Max Marks: 48/50

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.
CO2	Analyze public-key cryptosystem strategies using RSA and key exchange for strengthening information security.

Answer all the Questions

1. A security analyst at a small firm is asked to demonstrate classical encryption to new trainees. The plaintext "MEETATNOON" must be encrypted using a Caesar cipher with shift 7. Show the encryption and decryption process step by step, first converting each letter to its numeric value (A=0, B=1, ..., Z=25), applying the formulas $C = (P + 7) \bmod 26$ and $P = (C - 7) \bmod 26$, and then converting the numbers back to letters. Next, encrypt the same plaintext with a Vigenère cipher using the repeating key "SUN", showing the key shift for every letter position. Compare the Caesar cipher with the Vigenère cipher in terms of

   (a) key space (b) treatment of repeated patterns such as the double letters EE and OO and the repeated letters T and N, and

(c) vulnerability to cryptanalysis

Recommend one of the two only for a constrained educational demonstration, and justify your choice. Finally, explain why neither cipher is suitable for protecting modern confidential data, considering key size, determinism, lack of integrity protection and the strength of modern algorithms such as AES-256-GCM. **(10)**

(CO1) (BL3)

2. A cryptography instructor wants to show how matrix algebra is used in classical block ciphers.

Encrypt the plaintext "MATH" using a 2 × 2 Hill cipher with key K = [[3, 2], [5, 7]], using A=0, B=1, ..., Z=25. Show the conversion of the plaintext into column vectors (MA and TH), and show the matrix multiplication and modulo-26 reduction for each block to obtain the ciphertext. Determine whether the key matrix is invertible modulo 26 by computing its determinant and checking gcd(det, 26) = 1. Find the modular adjugate method, and verify that $K \cdot K^{-1} \equiv I \pmod{26}$. Explain how K^{-1} enables inverse of the determinant, compute the inverse key matrix K^{-1} using the decryption by applying it to the ciphertext blocks and recovering "MATH". Show why the alternative key K = [[2, 3], [4, 5]] is unacceptable by calculating its determinant and its gcd with 26. Finally, identify and explain a major weakness of the Hill cipher, such as its linearity and vulnerability to known-plaintext attacks.

(10) (CO1) (BL4)

3.a) A hospital network of 200 clinics must send large MRI archives (about 1 GB each) to a central radiology server, and every clinic must also exchange encryption keys with the server securely. The IT team is choosing between a symmetric cipher (AES-256-GCM) and the RSA public-key algorithm. Compare the two in terms of

(a) key usage (who holds which key), (b) performance and maximum data size per operation, (c) scalability, including the number of keys needed if all 200 clinics must communicate pairwise versus using RSA key pairs, and key-distribution requirements, including the need for certificates to authenticate public keys. **(5) (CO1) (BL4)**

b) Using this comparison, design a hybrid encryption workflow that uses each algorithm where it is most appropriate, listing the steps carried out by the sending clinic (certificate validation, DEK and nonce generation, AES-GCM encryption with associated data, RSA-OAEP key wrapping, and packaging) and by the receiving server (key unwrapping, decryption, and tag verification). Also explain how secure key storage should be handled (HSM/KMS, non-exportable keys, role-based access, MFA, audit logging, backup and recovery), and how key rotation should be handled (per-file DEKs, periodic rotation of the key-encryption

key, re-wrapping, retirement of old keys, and emergency rotation after suspected compromise).

(5) (CO2) (BL4)

4. For an RSA demonstration in a cryptography laboratory, let the primes be p = 11 and q = 19, with public exponent e = 7. Calculate the modulus n and Euler's totient $\phi(n)$, and confirm that gcd(e, $\phi(n)$) = 1. Determine the private exponent d such that $e \cdot d \equiv 1 \pmod{\phi(n)}$ using the extended Euclidean algorithm, showing every division step and the back-substitution. State the public key and the private key. Encrypt the message M = 5 using C = M^e mod n, showing the intermediate modular exponentiation steps. Verify the result by decrypting the ciphertext using M = C^d mod n (you may use the Chinese Remainder Theorem to simplify the calculation) and confirm that the original message is recovered. Finally, explain why real systems require much larger parameters (2048-bit or 3072-bit moduli) and careful padding such as RSA-OAEP for encryption and RSA-PSS for signatures, discussing factoring attacks, determinism, malleability and small-message attacks on textbook RSA.
(10) (CO2) (BL4)

5. A financial institution transmits confidential customer information across a network and must protect it against unauthorized access. Apply suitable classical encryption techniques, such as Caesar cipher, Playfair cipher, and Vigenère cipher, to encrypt and decrypt the given plaintext. Extend your solution by applying an appropriate symmetric or asymmetric cipher to secure the same information. Demonstrate the encryption and decryption procedures, compare the techniques in terms of key usage and security, and justify which approach is more suitable for protecting sensitive financial data. **(10) (CO1) (BL3)**

Continous Internal Assessment 1

N. Akshaya
192425129
CSA5101
Cryptography & Network security

1) Caesar Cipher and Vigenere Cipher

Given:

* plaintext = MEETATNOON
* Caesar Shift = 7
* vigenere key = SUN

A) Caesar Cipher

Formula:

$C = (p+7) \bmod 26$

Plaintext	M	E	E	T	A	T	N	O	O	N
Ciphertext	T	L	L	A	H	A	U	V	V	U

Encrypted text: TLLAHAUVVU

Decryption Formula: $P = (C-7) \bmod 26$

Recovered text: MEETATNOON

B) Vigenere Cipher

plaintext: MEETATNOON

Repeated key: SUNSUNSUNS

Formula:

$C = (p+k) \bmod 26$

Plaintext	M	E	E	T	A	T	N	O	O	N
key	S	U	N	S	U	N	S	U	N	S
Ciphertext	E	Y	R	L	U	G	F	I	B	F

Encrypted text: EYRLUGFIBF

Decrypted formula:

$$p = (C - k) \bmod 26$$

Recovered text: MEETATNOON

c) Comparison

Caesar	Vigenere
uses one Shift	uses multiple Shifts
Small key Space	larger key space
Easy to break	more difficult to break
Same letters give Some output	output depends on key positions

Recommendation: Vigenere is better for educational demonstrations because it uses multiple shifts.

Why not Suitable for modern security?

* Small key size
* Easy to attack
* Modern AES-256-GCM provides Stronger security

2) Hill Cipher

Given:

* plaintext = MATH

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

A = 0, B = 1, ... Z = 25.

A) Convert plaintext

M = 12, A = 0, T = 19, H = 7

$$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \qquad TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

B) Encryption

Formula:

$$C = Kp \bmod 26$$

For MA

$$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \\ 60 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

kI

For TH:

$$= \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 71 \\ \\ 144 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

To

Ciphertext : KITO

C) Check key Inveritibility

$Det(K) = (3 \times 7) - (2 \times 5)$

$= 21 - 10$

$= 11$

$gcd(11, 26) = 1$

∴ the key is invertible.

D) Find inverse matrix

$11^{-1} \bmod 26 = 19$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

E) Decryption

$$P = K^{-1} C \bmod 26$$

Applying the inverse matrix to KI and TO gives:

KI → MA

TO → TH

Recovered plaintext: MATH

F) Alternative key

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Det(K) = 10 - 12 = -2

gcd(-2, 26) = 2

Since GCD is not 1, the key is not invertible.

Weakness: Hill Cipher is linear and vulnerable to known-plaintext attacks.

Conclusion: Hill Cipher uses matrix multiplication for encryption and inverse matrices for decryption.

3) AES-256 - GCM and RSA

A) Comparison

Feature	AES-256-GCM	RSA
Type	Symmetric	Asymmetric
keys	Same secret key	public and private keys
Speed	Fast	slow
usage	Large files	Small data / key protection
Security	Confidentiality and Integrity	public-key security

Key usage:

* AES uses the same key for encryption and decryption.

* RSA uses public and private keys.

performance:

* AES is suitable for 1 GB MRI files.

* RSA is not suitable for large-file encryption.

Scalability:

For 200 Clinics: $\frac{200 \times 199}{2} = 19900$

pairwise AES Communication requires 19,900 Shared keys.

RSA requires 201 key pairs (200 Clincies and one server)

Key distribution:

* AES needs Secure key sharing
* RSA uses digital certificates to verify public keys.

B) Hybrid Encryption

Sending Clinic:

1. Validate Server Certificate.
2. Generate AES key (DEK) and nonce.
3. Encrypt MRI files using AES-256-GCM.
4. Wrap the DEK using RSA public key with RSA-OAEP.
5. Send encrypted file, DEK, nonce and authentication tag.

Receiving Server:

1. Receive encrypted data.

2. unwrap DEK using RSA private key.

3. Decrypt using AES-256-GCM.

4. Verify authentication tag.

Secure key storage:

* use HSM/KMS.

* use non-exportable keys.

Key rotation:

* use Separate DEKs for files.

* Rotate encryption keys periodically.

* Re-wrap DEKs.

* Retire old keys.

Conclusion: AES encrypts large files, while RSA protects AES keys. Both provide an efficient hybrid Solution.

4) RSA Encryption and Decryption

Given

p: 11, q: 19, e: 7, M: 5

A) Calculate modulus

$n = p \times q$

$n = 11 \times 19$

$n = 209$

B) Calculate Totient

$\phi(n) = (p-1)(q-1)$

$= 10 \times 18$

$= 180$

C) Check GCD

$gcd(7, 180) = 1$

$\therefore$ e is valid

D) Find private key

$7d \equiv 1 \mod 180$

Euclidean algorithm:

$180 = 25(7) + 5$

$7 = 1(5) + 2$

$5 = 2(2) + 1$

Back Substitution:

$1 = 5 - 2(7 - 5)$

$= 3(5) - 2(7)$

$= 3(180) - 77(7)$

$d = -77 \bmod 180$

$d = 103$

E) keys

Public key : (7, 209)

private key : (103, 209)

F) Encryption

$C = M^e \bmod n$

$C = 5^7 \bmod 29$

$C = 168$

G) Decryption

$M = C^d \bmod n$

$M = 168^{103} \bmod 209$

using modular exponentiation : $M = 5$

Recovered plaintext = 5

H) Why large RSA keys are required ?

1. Small keys are easy to break.
2. Factoring attacks can recover private keys.

Conclusion : RSA successfully encrypts & decrypts the given message.

5) Classical and Modern Encryption

Given:

* plaintext = HELLO
* Caesar key = 3
* playfair key = MONARCHY
* vigenere key = KEY

A) Caesar Cipher

Formula:

$$C = (P + 3) \bmod 26$$

HELLO → KHOOR

Decryption:

$$P = (C - 3) \bmod 26$$

KHOOR → HELLO

B) playfair Cipher

key: MONARCHY

Matrix

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

plaintext : HELLO

pairs = HE LX LO

Encryption :

* HE → CF

* LX → SU

* LO → PM

Decryption gives HELLO after removing the extra X.

C) vigenere Cipher

Plaintext : HELLO

Key : KEYKE

Formula : C = (P + K) mod 26

plaintext	H	E	L	L	O
Key	K	E	Y	K	E
Ciphertext	R	I	J	V	S

Ciphertext : RIJVS

Decryption gives HELLO

D) AES-256-GCM

AES-256-GCM is a modern symmetric encryption algorithm.

Encryption:

1. Generates a 256-bit secret key.
2. Generates a unique nonce.
3. Give plaintext HELLO as input.
4. Encrypt using AES-256-GCM.
5. Generate ciphertext and authentication tag.

Decryption:

1. Receive ciphertext
2. use the same key and nonce.
3. Decrypt the data.
4. Verify authentication tag.
5. Recover HELLO

E) Comparison

Algorithm	Key	Security
Caesar	Shift	Very week
playfair	Matrix	weak
vigenere	Key word	weak
AES-256-GCM	256-bit key	Strong

Conclusion: Classical ciphers are useful for learning but AES-256-GCM is suitable for modern data.